

MAGI: state of validation

An internal reference for paper writing

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Status of this document

This is an internal working reference, not a manuscript and not submission-ready. It exists so that numbers entering the proceeding and the flash talk can be traced to the run that produced them.

Every numeric claim carries a provenance tag [Ex] resolving to Table 4. Tags point at files and runs on the local machine; they are *not* literature citations. This document contains no external references, because none have been verified for it.

Superseded and withdrawn results are retained and marked, not deleted. Work in progress is marked as such and carries no result.

1 Scope

MAGI is a conditional variational autoencoder that replaces Geant4 transport from a spacecraft or laboratory mass model down to a closed surface (the “CryoSphere”) surrounding a cryogenic detector. Instead of tracking primaries through the full assembly, MAGI generates particle crossings directly on that surface; those crossings are then injected into the same mass model and transported the short remaining distance.

The validation question is therefore narrow and well posed: *does a MAGI crossing produce detector events at the same rate and with the same spectrum as a real recorded crossing, in the same geometry?*

Three independent sources have been tested: cosmic rays on the SRON X-IFU model (CR), cosmic rays on the DM1.2 laboratory cryostat, and ^{40}K radioactivity on the SRON model.

1.1 Model version under validation

All results below are **v0.8.2**, class `CVAE_MixEnergy_ContPhi_TaskAdaptive`: conditional rational-quadratic spline normalizing-flow continuum with CDF pre-warp, pinned 4 eV emission lines, focal-gated line/continuum supervision, and a learnable conditional coupling prior with zone conditioning. `latent_dim=8`.

v0.8.3 (`energy_condition_geometry=True`, `latent_dim=12`) was implemented and trained at three seeds and is **not adopted** (§10).

2 The normalization-free comparison

The central method, used for every headline number here:

$$R = \frac{N_{\text{events}}^{\text{MAGI}} / N_{\text{crossings}}^{\text{MAGI}}}{N_{\text{events}}^{\text{real}} / N_{\text{crossings}}^{\text{real}}} \quad (1)$$

Both sides are counted *per injected crossing*, through the identical mass model, binary, physics list and scoring code. R requires no exposure, no detector area, no flux normalization, and—critically—no crossing efficiency χ . It is insensitive to every quantity that has historically caused discordance between MAGI-side and Geant4-side bookkeeping.

The *efficiency factor* quoted in some tables is a different normalization, retained for continuity with earlier analysis:

$$\varepsilon = \frac{N_{\text{counts}}^{\text{MAGI}}/N_{\text{shot}}^{\text{MAGI}}}{N_{\text{counts}}^{\text{ref}}/N_{\text{shot}}^{\text{ref}}}, \quad R = \varepsilon \times \chi, \quad (2)$$

where MAGI’s shot particles are CryoSphere crossings and the reference’s are thrown primaries. A perfect surrogate returns $\varepsilon = 1/\chi$, not 1.

3 Result 1 — Cosmic rays, DM1.2 laboratory cryostat

The cleanest measurement available, and the only one with a multi-seed ensemble at detector level.

Configuration. Reference: 2×10^8 muons thrown isotropically (`/gps/ang/type iso, mintheta 0, maxtheta 90`) through the DM1.2 mass model, yielding 999,358 CryoSphere crossings and 1,733 detector events [E1]. MAGI: 1,200,000 crossings generated from a 3-seed ensemble (seeds 42, 7, 13; two files per seed) and injected into the same model, yielding 2,019 detector events [E2]. Sphere: centre (0, 0, 5) mm, $R = 95$ mm, verified from model metadata at generation time [E3].

Table 1: DM1.2: counts per injected crossing, MAGI vs full simulation.

Band	Reference	MAGI	R
all deposits	1,733	2,019	0.970 ± 0.032
100–300 keV (MIP)	1,325	1,589	0.999 ± 0.037
300–1000 keV	310	323	0.868 ± 0.069 [E2]
30–100 keV	42	46	0.912 ± 0.195
0.1–30 keV	41	43	0.873 ± 0.191
>1000 keV	13	17	1.089 ± 0.401

Efficiency factor. $\chi_{\text{DM1.2}} = 999,358/200,000,000 = 4.9968 \times 10^{-3}$, so a perfect surrogate returns $1/\chi = 200.13$ [E2].

Table 2: DM1.2 efficiency factor. The 0.1–30 keV row is the one directly comparable with the SRON CR value in §4.

Band	ε	ideal $1/\chi$	$R = \varepsilon\chi$
0.1–30 keV	174.8 ± 38.2	200.13	0.873
1–20 keV	217.9 ± 56.8	200.13	1.089 [E2]
100–300 keV (MIP)	199.9 ± 7.4	200.13	0.999
all deposits	194.2 ± 6.4	200.13	0.970

What can and cannot be claimed. **DM127 is an anti-coincidence detector** (ACDSD in `GDMLDetectorConstruction.cc`), so the X-IFU 0.1–30 keV science band is *not* the relevant band for it and must not be labelled as such — the 0.1–30 keV row is retained only because it is the band the SRON efficiency table uses. The physically relevant band here is the MIP peak, and that is also where the statistics are: 1,325 reference events, $\pm 2.7\%$, with ε within 0.4% of ideal. The 0.1–30 keV row is an *integrated rate only* — 41 reference events, $\pm 15.6\%$ before MAGI contributes any error. More MAGI generation cannot improve it; the reference is the cap.

DM1.2 has a **single** sensitive detector, so there is no across-detector spread to quote as a systematic; the ± 38.2 is pure Poisson.

Structural checks (all pass). Crossing coverage $1,733/1,733 = 100.0\%$ [E2]; global species fractions correct to 0.3% (γ 1.003, e^- 1.002, e^+ 1.039, μ^- 0.997) [E2]; impact-parameter distribution 0.912–1.010 over $b < 1$ to $b < 50$ mm [E2].

Known defect: a coherent 4 % deposit shift. A two-sample Kolmogorov–Smirnov test on all deposits gives $D = 0.0558$, $p = 0.0057$ [E2]. This is *not* a shape distortion but a uniform downward shift: deposit quantiles 25–95 % run 0.984, 0.960, 0.959, 0.940, 0.962, and the maximum CDF separation falls on the MIP peak at 186 keV. It localises to the MIP band ($D = 0.0594$, $p = 0.012$); the 300–1000 keV band and everything below 100 keV are indistinguishable ($p = 0.81$ and 0.99).

The cause is path length. Among detector-aimed crossings MAGI’s tracks are too close to normal incidence: median $\sec\theta$ is 0.955 of the reference value, against a measured deposit-median ratio of 0.960 [E4]. Underlying it is a *direction-conditional species-mix error*: at $b < 10$ mm MAGI runs +8.8 % muons, –13.6 % e^- , –7.7 % γ , while global fractions are correct to 0.3% [E4]. Per-species energies are correct (muons 0.99–1.03 at every cut).

Prerequisite finding: a geometry overlap that destroyed 39 % of crossings. At $R = 105$ mm the CryoSphere intersected PlateCu1 (a 150 mm-radius, 5 mm-thick copper disc at $z = 106.5$, spanning $z \in [104, 109]$) by at least 1.29 mm — $13\times$ the shell thickness — and was placed with overlap checking disabled. Geant4 navigation inside an overlap is undefined, and 39.4 % of detector events had *no recorded crossing*, uniformly 58–63 % across five independent runs, 93.5 % of them primary-muon deposits [E5]. Moving the sphere to $R = 95$ mm restores 100.0 % coverage. The SRON sphere was checked and is clean.

Separately, the DM1.2 muon macro had used `/gps/ang/type` **focused** ten times, aiming every muon at the origin (median $\cos$ 0.994 to the origin, 56.8 % inside $\cos > 0.99$); absolute rates from that configuration were never physical [E5]. All DM1.2 results here are from the `iso` rerun.

4 Result 2 — Cosmic rays, SRON X-IFU model

Normalization-free. $R = 0.848 \pm 0.125$ [E6], from 51 counts in 1,600,000 injected real crossings against 500 counts in 18,500,000 injected MAGI crossings, same geometry and binary. (The uncertainty combines both arms; an earlier statement of ± 0.14 used the reference arm alone.)

Per-detector flux ratio against the RUN CRYOAC reference, 1–20 keV: 0.76, 0.90, 0.85, 0.57, 0.81, 0.87 over detectors 1–6, mean **0.79** [E7]. Spectral shape is **indistinguishable** from the reference: KS $D = 0.048$, $p = 0.20$ [E7]. For contrast, v0.6.2 gave $p = 0.0014$.

Efficiency factor. Counts in 0.1–30 keV [E7]:

Normalized, $R = \varepsilon\chi = \mathbf{0.76}$, consistent with the 0.79 of the flux ratio and the 0.848 of the normalization-free arm, by three independent code paths.

The SRON–DM1.2 contrast. This is the most important open physics question in the validation:

	ε (0.1–30 keV)	ideal $1/\chi$	R
SRON CR, 6 detectors	124.1 ± 15.2	164.4	0.76
DM1.2, single detector	174.8 ± 38.2	200.13	0.873 ± 0.191
DM1.2, MIP band	199.9 ± 7.4	200.13	0.999 ± 0.037

Table 3: SRON CR efficiency factor per detector, v0.8.2. Ideal value $1/\chi_{\text{CR}} = 164.4$.

Detector	Reference counts	ε (v0.8.2)
1 — TES array	28,402	124.8 ± 5.6
2 — Back CryoAC	11,187	142.6 ± 9.6
3 — Lateral	2,922	121.3 ± 17.3
4 — Lateral	3,036	98.1 ± 15.2
5 — Lateral	2,829	122.8 ± 17.7
6 — Lateral	3,039	135.3 ± 17.9
mean		124.1
std across detectors		15.2

DM1.2 sits at its ideal value where statistics are good; SRON shows a $\sim 24\%$ deficit that DM1.2 does not reproduce. The DM1.2 overlap bug was excluded as the cause — the SRON sphere does not overlap.

Identified contribution to the SRON deficit. An **inconsistent ingoing-particle cut** [E8]. `EventAction.cc` skips outgoing crossings at registration, but the SRON training sets predate that code: the CR training set contains 10.37% outgoing crossings and the ^{40}K set 22.53% , while the DM1.2 set contains 0.00% . MAGI applies no ingoing/outgoing selection either, and generates 13.59% outgoing against the training set’s 10.37% — accounting for 3.6% of the rate deficit. This also means **SRON and DM1.2 models are on different conventions**.

5 Result 3 — ^{40}K , SRON BuildingModel

Correction 17/08/2026: the training population is albedo, not the crossing population

The ^{40}K training set was recorded by Old-Simulations/Sim/S1GDMLSRON_CryoSphereEmission, whose EventAction.cc:237 carries SphereCenter(0., 0., -3007.6*mm) — the 2500 mm error of §4. 100.00 % of its crossings are ingoing with respect to that wrong centre (and 97.28 % move downward), against 77.45 % ingoing w.r.t. the true -507.66. ^{40}K gammas travel *up* from the floor, so a cut toward a centre 2500 mm *below* kept only photons that passed the sphere, scattered off the structure above and came back.

The spectrum confirms it independently: the median is **211.2 keV** against the Compton 180° backscatter energy of the 1460.8 keV line, **217.5 keV**; 30.4 % of crossings lie within $\pm 15\%$ of it, the histogram peaks at 160–180 keV, and the 1461 keV line appears *once in 1.87M* — an unscattered line photon cannot be moving downward at the sphere.

Geometry was excluded by measurement, not assumed: an A/B gave 365.6 keV (original slab), 374.7 keV (reduced slab) and 366.8 keV (shielded BuildingModel), all ~ 370 keV and insensitive to the slab at the 3 % level. Re-crossings (0.86 % of events; first-crossing-only median unchanged at 211.2 keV) and species composition (99.3 % gamma both sides) were excluded too.

Ra-226 and Th-232 are affected identically and cannot substitute: both are 100.00 % ingoing w.r.t. the buggy centre, both have medians at 0.83–0.86 of their backscatter energies, Ra-226 has *zero* events at its 1764.5 keV line, and Th-232’s maximum recorded energy (2505 keV) falls *below* its 2614.5 keV line altogether.

What survives. The numbers below are unchanged and remain a valid *closure* result — MAGI reproduces the distribution it was trained on, and that survives transport to detector level. They are **not** evidence about ^{40}K background physics, because both arms of the comparison draw on the same biased population. Restated as a physics claim they would be wrong.

$R = 1.118 \pm 0.373$, **KS** $D = 0.2384$, $p = 0.60$ [E9]: 2×10^8 injected on both sides into SRON_CCNCoolerwithXFDM_noShields_BuildingModel.gdml, giving 19 MAGI counts (9.50×10^{-8} per crossing) against 17 from real recorded crossings (8.50×10^{-8}). Matched N by construction. **^{40}K passes at detector level.**

The real-crossing arm recycles 1,873,609 recorded crossings $\sim 107\times$, which is sound because Geant4 re-randomizes the interaction on each replay; continuity was verified from the crossing counts ($\sim 40.15\text{M}$ re-registered per job, 0.80 per injected event, uniform across four jobs) rather than assumed [E9].

The 22× gap — superseded explanation. Against the 44-days reference the same run reads 0.049 ± 0.012 . This was previously attributed [E10] to the two runs not sharing a ^{40}K source volume (a $57\times$ ratio between a $2000 \times 2000 \times 400$ mm box at $z = -1600$ mm and $750 \times 750 \times 50$ mm at $z = -1250$ mm).

That attribution is now known to be incomplete, and its direction was wrong. The user confirms the slab was the *reduced* one and the model unshielded; the old campaign’s own macro samples the large box confined to the reduced slab, so both runs sample the same physical volume and the $57\times$ never applied as stated. The dominant effect is the albedo selection above: the training set retains only the back-scattered fraction, which both suppresses the yield and softens the spectrum. The two still cannot be chained, and the conclusion — MAGI sits with the population it was trained on — stands, but for a different and larger reason. **^{40}K returns in September (§14);** this section is not to be used as a physics result before then.

Bookkeeping correction. NgenPotassiumForMAGI_v082 was declared 580,000,000; measured from the run’s own output it is 538,205,667 [E11]. The output file concatenates two jobs, with exactly one backward `EventId` jump at row 398,400; estimating each job separately by $N = \max \cdot (n + 1)/n$ gives 438,145,967 and 100,059,700. The declared value is 7.8 % high and outside the 95 % region of both jobs’ order statistics.

The same estimator was run as a control on two other declared counts and *confirms* them (^{40}K v0.7.2 ratio 1.0001; CR v0.8.2 ratio 0.9998), so it is not systematically low. Every v0.8.2 ^{40}K *Flower* flux and efficiency rises by 1.0777; the Detector-1 efficiency goes 3472 $\rightarrow$ 3741. The detector-level result above is unaffected — it never uses the constant.

6 Figures for the proceeding

Built by `paper_figures/make_fig1.py` and `make_fig2.py` [E16]; vector PDF at 7.1 in width, sized for a two-column page.

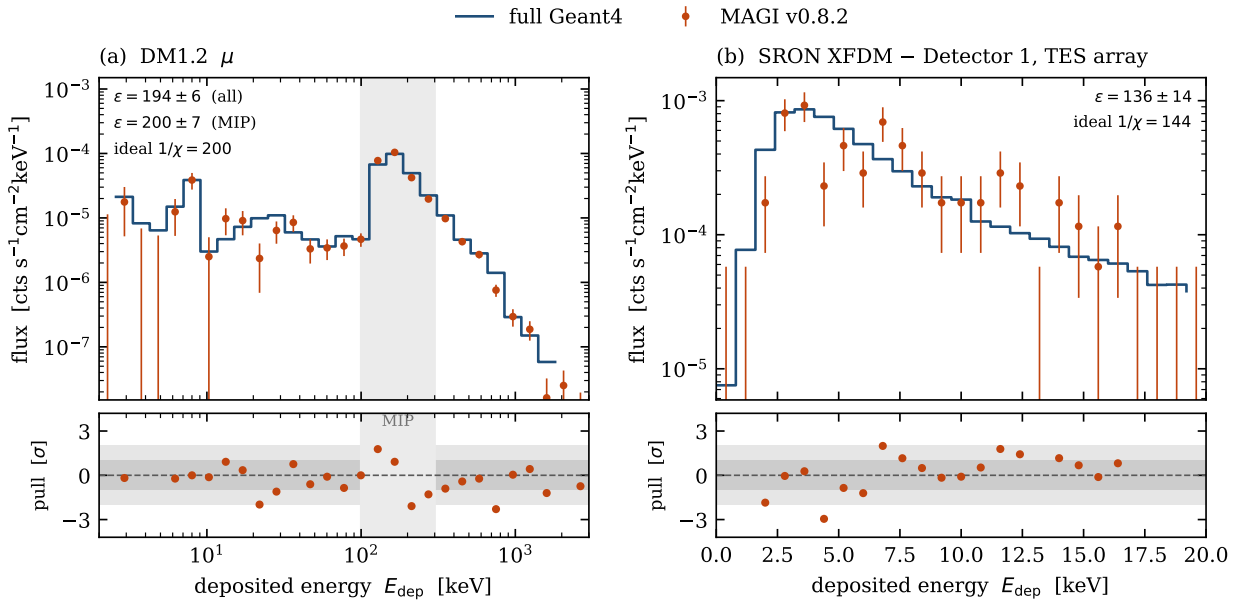


Figure 1: **Figure 1 — spectral comparison.** (a) DM1.2, full deposited-energy band, normalized differential rate. The MIP band is shaded. (b) SRON XFDM Detector 1 (TES array), flux against the RUN CRYOAC reference over 0–20 keV. Ratio panels carry *exact* Poisson confidence intervals (Clopper–Pearson on the conditional binomial), not $\sqrt{N}$ propagation — several bins hold single-digit counts where the true interval is strongly asymmetric. **Caption must state** that the two panels sit on different flux references: (a) uses `Sim/OmniFluxes.tsv` with $R_{\text{gen}} = 118\text{ cm}$, (b) uses `Sim/S1GDMLSRON/OmniFluxes_Run1.tsv` with the Building radius and its scale factors. The y -axes are not comparable between panels.

Why DM1.2 and not the others. Three sources have a loadable v0.8.2 checkpoint (Torio and Radio do not — their checkpoints on this machine are incomplete, weights still on Drive). **DM1.2 was chosen**, and the alternatives are recorded here because the choice is not the one the summary statistic alone would make.

Dataset	mean $ \Delta\rho $	max $ \Delta\rho $	strongest $ \rho $ in the truth
^{40}K (Small) [†]	0.0134	0.0266	0.21
CR	0.0149	0.0417	0.28
DM1.2 iso	0.0176	0.0836	0.62

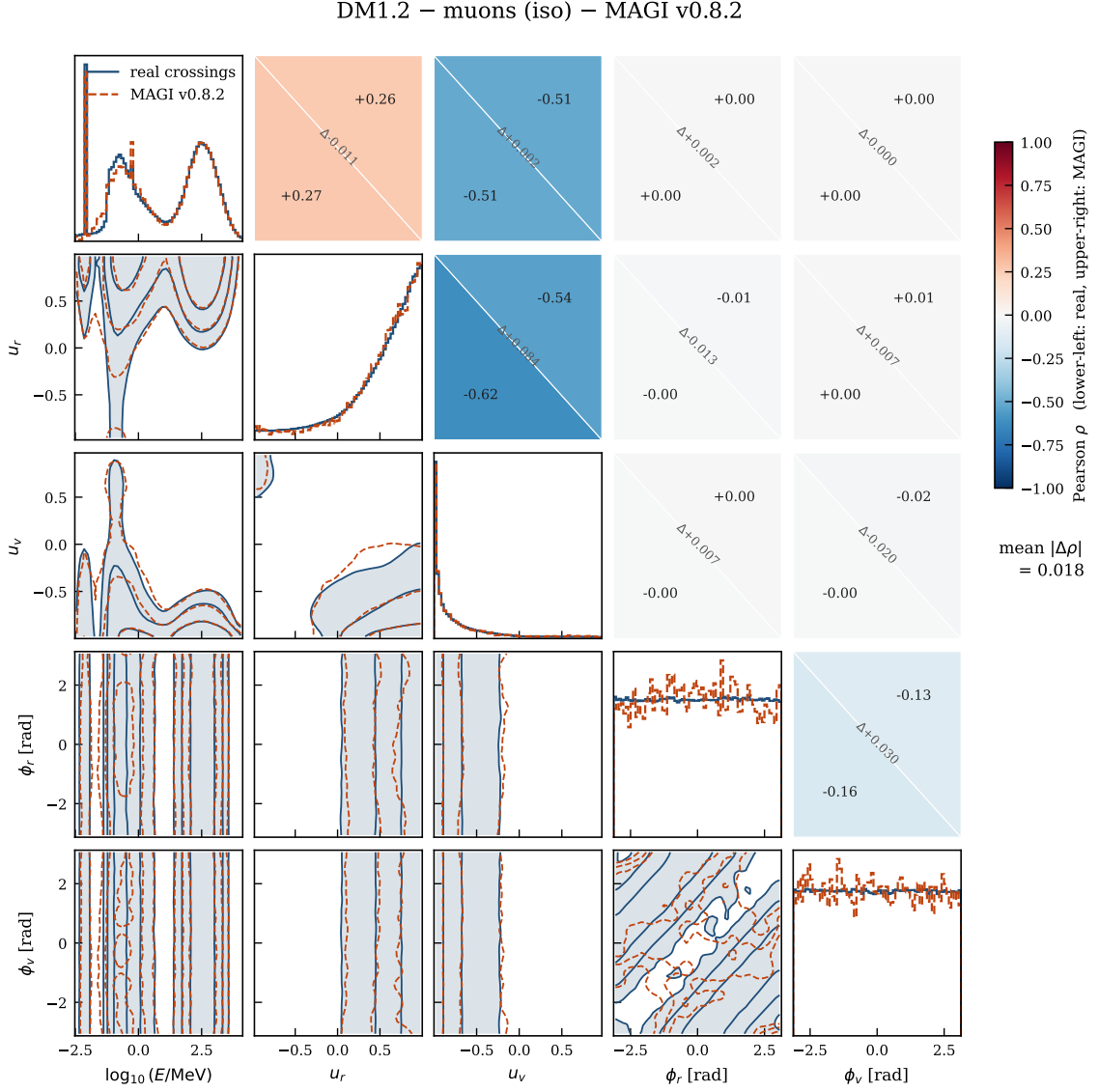


Figure 2: **Figure 2 — DM1.2.** Crossing population, real vs MAGI. Lower triangle: 2D contours. Diagonal: marginals. Upper triangle: each cell split diagonally, lower-left half the real Pearson ρ , upper-right half MAGI's, on a common scale with the residual printed — a visually uniform cell is a reproduced correlation.

[†] Measured on the albedo population of §5; a valid fit-quality metric on that set, not a statement about ⁴⁰K physics. The same caveat applies to Torio and Radio.

DM1.2 has the *worst* mean residual of the three, and is still the right figure. ⁴⁰K wins on the number, but every correlation in its truth is $|\rho| \leq 0.21$ — there is almost no structure to reproduce, and a referee can fairly say that reproducing near-zero correlations is not a test. DM1.2 carries genuinely strong structure ($u_r - u_v = -0.62$, $\log E - u_v = -0.51$) and MAGI tracks it. The honest cost is its one visible miss: $u_r - u_v$ reads -0.62 real against -0.54 MAGI, $\Delta = +0.084$, the largest single residual anywhere. That is worth showing rather than hiding — it is the same direction-conditional geometry error that §9 identifies, visible in the one panel where the truth has enough structure to expose it.

7 Result 5 — the CR retest and the memorisation test

7.1 The ingoing-cut prediction, scored out of sample

The corrected CR training set (3,737,351 crossings, $\chi = 6.921 \times 10^{-3}$, 0.00% outgoing) was used to retrain v0.8.2 on Colab. Before either model was injected, a prediction was recorded from the training sets alone: aimed yield per primary at $b < 20$ mm $\times 1.268$ overall and $\times 1.092$ for muons.

Both checkpoints were then generated and injected in the same session, same geometry (**Flower_fixed**), matched N , both emitted on the corrected sphere. **Only the training set differs.**

Band	per crossing	per primary	predicted	σ
all deposits	1.097	1.248 $\pm$ 0.047	1.268	0.4
0.1–30 keV	1.053	1.198 $\pm$ 0.121	1.268	0.6
1–20 keV	1.026	1.167 $\pm$ 0.130	1.268	0.8
1–7 keV (MIP)	0.876	0.996 $\pm$ 0.149	1.092	0.6

A units trap, recorded because it nearly inverted the conclusion

The first reading of this table used the *per injected crossing* column and declared the prediction falsified at 4.2σ . That column divides χ out — and χ is exactly what the ingoing fix changed ($6.084 \times 10^{-3} \rightarrow 6.921 \times 10^{-3}$). The prediction was stated per *primary*, so the correct comparison multiplies by χ . It agrees to 0.4σ . Both columns are meaningful and neither is wrong: per crossing measures how efficiently a MAGI crossing converts into a detector event, per primary is the flux. Quote the one the claim is about.

Against the full simulation. Detector 1, flux units, normalisation identical to Fig. 1(b) (Fig. 3):

Checkpoint	MAGI/full, 0–20 keV	from unity
censored training set	0.802 $\pm$ 0.078	2.6 σ
ingoing-fixed training set	0.963 $\pm$ 0.043	0.85 σ

The deficit moves from 2.6σ below unity to consistent with it. The *improvement* itself is $+0.161 \pm 0.089$, **1.8 σ** .

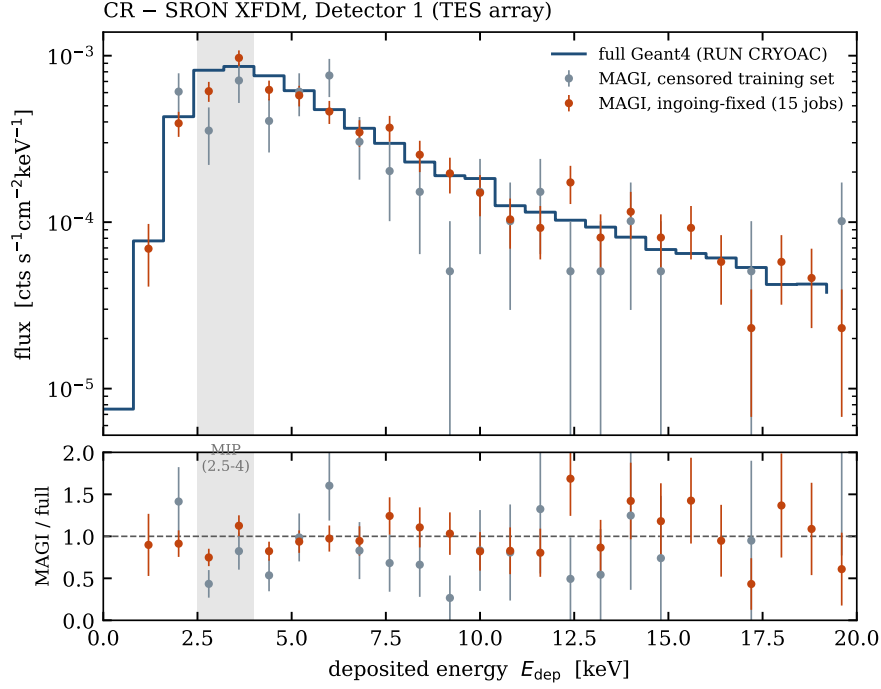


Figure 3: CR at detector level, SRON XFDM Detector 1. Full Geant4 (RUN CRYOAC) against MAGI trained on the censored (3 jobs) and on the ingoing-fixed set (15 jobs, pooled from `cr_retest` and `cr_retest_morestat`), in flux units with the Fig. 1(b) normalisation. Each MAGI crossing stands for $1/\chi$ primaries, and χ differs between the two checkpoints — which is the effect under test. Shaded band: MIP, 2.5–4 keV.

Statistics campaign, 20–21/08/2026: 114 → 566 Detector-1 events

The number above superseded 0.941 ± 0.088 (107 and 114 Detector-1 events, 1.2σ improvement, “suggestive, not decisive” – Open Items §14). `cr_retest_morestat` [E22] added 12 more “new”-arm-only jobs (χ -matched, same geometry, same radius gate, batches of 4 to match the machine’s 4-P-core throughput) on top of the original 3, pooled for $4.5\times$ the crossings and $\approx 5\times$ the Detector-1 events. Error dropped $0.088 \rightarrow 0.043$, tracking $\sqrt{N}$ throughout three intermediate readings (0.941, then 1.031, 0.955, 0.963 as statistics accumulated) — consistent movement, not drift. The “old” (censored) arm was deliberately **not** extended: only the ingoing-fixed number needed tightening for the paper.

MIP band (2.5–4 keV): $R = 0.943 \pm 0.083$ (130 events), against the censored checkpoint’s 0.634 in the same band — the ingoing fix’s improvement is visible in this specific window too, not only in the 0–20 keV integral.

A methodological note worth keeping: an early re-check of this pooled result, run mid-campaign while the final batch was still injecting, returned $R = 0.734$, 3.4σ from unity — looking like a real, significant finding. It was a counting bug: a job directory can exist (`mkdir`’d at injection start) with a real injected-crossing count on disk before its `proc/` output does, and the pooling function was counting such a job’s crossings in the denominator while its (not-yet-existing) events contributed nothing to the numerator — silently deflating R . Fixed by gating each job on its processed output actually existing before counting it at all; re-verified against the already-confirmed 0–20 keV number before trusting the fix for anything else.

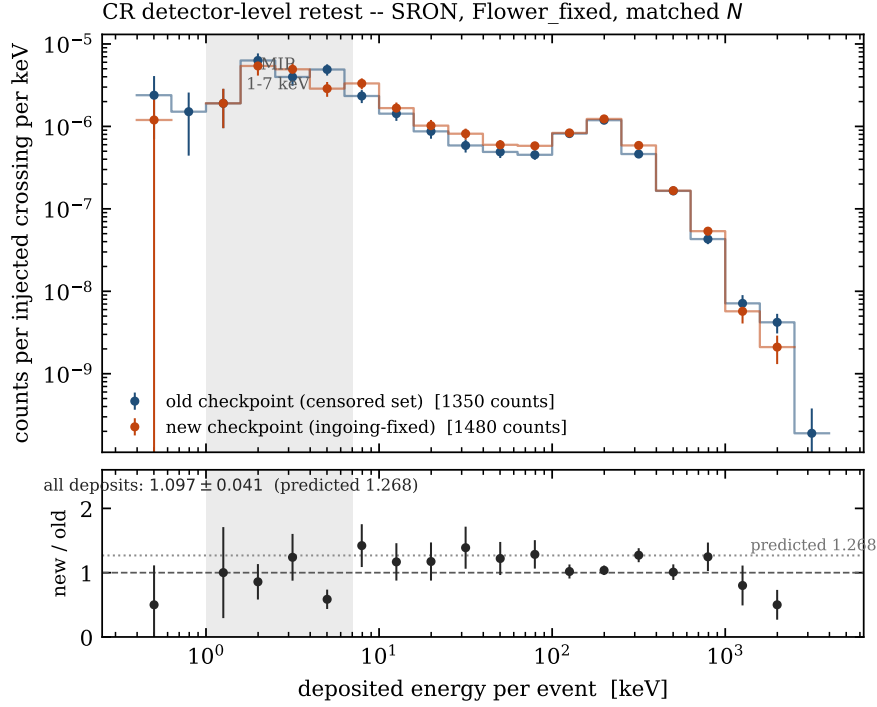


Figure 4: The same two checkpoints against each other, per injected crossing, over the full deposit range. This is the χ -free view: it isolates how efficiently a crossing converts into a detector event, and is flat near unity apart from the MIP band.

7.2 Arm A — the corrected checkpoint against real crossings

The retest above was model against model. This is model against *truth*: 3,583,201 real recorded crossings and 3,590,382 MAGI crossings, injected back to back into the same geometry, same binary, same physics, both as text through the same parser, both on the corrected sphere.

Band	real rate	MAGI rate	R
all deposits	4.382×10^{-4}	4.033×10^{-4}	0.920 ± 0.034
0.1–30 keV (X-IFU)	5.665×10^{-5}	4.791×10^{-5}	0.846 ± 0.088
1–20 keV	4.940×10^{-5}	3.788×10^{-5}	0.767 ± 0.087
1–7 keV (MIP)	2.623×10^{-5}	2.117×10^{-5}	0.807 ± 0.124

The old checkpoint scored 0.848 ± 0.125 in the X-IFU band. The corrected one scores 0.846 ± 0.088 — **the same number**, on 3.6×10^6 crossings per arm instead of the 51 counts behind the older value.

The three runs are mutually consistent. From the retest, new/old per crossing is 1.053 in this band; with new/real = 0.846 that implies old/real = 0.803, against the published 0.848 ± 0.125 . Independent routes, one answer.

Two different quantities, both now measured

Per primary (flux): 0.941 ± 0.088 of the full simulation. The ingoing fix repaired χ , so the *normalisation* is right.

Per crossing (conversion): 0.85–0.92. A MAGI crossing still yields 8–15 % fewer detector events than a real one.

These do not conflict. The flux problem is solved; a **conversion deficit** survives, and it lives in the model, not in the censored training set.

What the deficit is not. Three candidates were excluded by direct measurement on 1.2×10^6 real and 1.2×10^6 generated crossings:

Quantity	MAGI/real	verdict
aimed crossings, $b < 20$ mm	1.021 ± 0.006	not the cause
aimed crossings, $b < 10$ mm	1.012 ± 0.013	not the cause
cos(angle to detector), 99.9 % tail	0.99930 vs 0.99933	not the cause
species fraction, μ^-	1.000	not the cause
species shares among aimed	within 1 %	not the cause

MAGI points *as many* crossings at the detector, with the same species mix. The angular tail that matters is reproduced to the fourth decimal.

What remains. Two residuals, neither yet shown to be the cause:

Among aimed crossings ($b < 20$)	real	MAGI	ratio
median E , γ	423.0 keV	445.1 keV	1.052
median E , μ^-	819.4 MeV	784.4 MeV	0.957
median E , e^-	2.418 MeV	2.928 MeV	1.211
μ^- share of aimed	0.4105	0.3973	0.968
broad forward population, $b < 100$ mm	0.8155	0.7619	0.934

The $b < 100$ mm figure is suggestive — 0.934 against an all-deposits R of 0.920 — but a 6.6 % shortfall in the wide forward cone alongside a *surplus* in the core is a reshaping of the angular distribution, not a simple loss, and it is not established that it drives the detector rate.

The mechanism is not localised. It is not normalisation, not aiming, not species. It is a few-percent, species-dependent distortion of the joint distribution whose effect on a small detector is an order of magnitude larger than the distortion itself — which is exactly the regime a marginal-based check cannot see, and why the downstream test exists.

7.3 Localising the deficit: a classifier two-sample test

Real and generated crossings are separable. A gradient-boosted classifier trained to tell real from MAGI crossings, on the seven continuous features plus species, reaches **AUC** = 0.68 (0.5 would be indistinguishable). This is a sharper instrument than anything used before: it sees the joint, it is cheap, and on 4×10^5 samples per class its statistical error is negligible next to the ± 0.09 on R .

The mismatch is diffuse, not a broken head. Permutation importance is flat across all six continuous variables and essentially zero on species:

feature	ΔAUC	feature	ΔAUC
$\sin \phi_v$	+0.0724	$\log_{10} E$	+0.0556
$\sin \phi_r$	+0.0702	u_r	+0.0508
u_v	+0.0633	is_gamma	+0.0049
$\cos \phi_v$	+0.0624	is_μ [−]	+0.0008
$\cos \phi_r$	+0.0622	is_e [−]	+0.0003

A wrong marginal would put one variable far above the rest. Instead the classifier separates only by reading the *combination*. Per-species AUCs agree (γ 0.651, μ^- 0.614, e^- 0.630), so it is not one population. This is consistent with everything else measured: marginals reproduced to 5–11 % RMS, correlation residual 0.0147 against KDSources’s 0.0025, and DM1.2 over-dispersed at 1.536 in nearest-neighbour distance (§7.4).

Reweighting cannot repair it — and that is the result. The planned test was to reweight MAGI to the real joint by likelihood ratio and re-inject. It was run twice and *refused to inject both times*, because the reweighted sample stayed separable:

	AUC before	AUC after
single-shot, uncalibrated, 2:1 pool	0.672	0.608
isotonic-calibrated, 4 iterations, 6:1 pool	0.682	0.577

The second attempt reached in-loop AUC 0.537 but validated at 0.577, with maximum weight climbing 11 → 52 and effective sample size falling to 5.4 M of 12 M. That is weight degeneracy: **you cannot upweight samples the generator never produced**. The obstruction is coverage, not density shape.

Where MAGI does not go. With $c(x) = P(\text{real} | x)$:

	% of real	% of MAGI	density ratio
$c > 0.70$	7.62 %	1.43 %	5.3×
$c > 0.80$	2.73 %	0.28 %	9.9×
$c > 0.90$	0.62 %	0.024 %	25.8×
$c > 0.95$	0.10 %	0.001 %	104×

The under-covered events ($c > 0.8$) are 87.4 % gamma against 56.9 % overall, with median energy **511.0 keV** against 2831 keV, median $u_r = +0.108$ against +0.326 and median $v_z = -0.004$ against -0.427 : **annihilation photons entering horizontally near the sphere’s equator**. The model carries exactly one pinned line, at $10^{-0.2916} = 511$ keV. It has the line and it has the angular distribution; what it lacks is their joint. The energy head and the geometry heads are coupled only through an 8-dimensional latent, and this combination falls through it.

This hole does not explain the detector deficit

The under-covered events are *depleted* among aimed crossings — enrichment 0.87×, carrying 2.38 % of the aimed population against 2.73 % of the whole. The most prominent mismatch is not where the detector looks. The conversion deficit is therefore characterised (diffuse, joint, coverage-flavoured, resistant to reweighting) but **not mechanistically explained**. Do not present the 511 keV hole as its cause.

What this unlocks. Every architecture change until now has been judged on marginals, or on a downstream R carrying ± 0.09 . The classifier AUC is sensitive to the joint, costs minutes, and resolves differences far below what R can. **energy_condition_geometry** — conditioning the energy head on sampled geometry, the exact mechanism this defect calls for — already exists as the v0.8.3 feature that was implemented, tested and not adopted (§10). It had never been evaluated against this metric. That test was brought forward from September and launched 19/08; see §11.

7.4 Memorisation — does MAGI generate, or recall?

The KDSOURCE benchmark showed a $287\times$ amplification does not by itself distinguish MAGI from a kernel resampler. This test does. For each generated sample, the nearest-neighbour distance to a reference set of real crossings, in the model’s own feature space with the azimuths carried as (cos, sin) pairs so the metric respects periodicity. Memorisation puts generated samples closer to training points than real data lies from itself.

Arm	DM1.2		CR (ingoing-fixed)	
	ratio	near-dup	ratio	near-dup
real $\rightarrow$ real (LOO)	1.000	1.00 %	1.000	1.00 %
held-out real (<i>control</i>)	0.999	1.07 %	1.001	0.90 %
MAGI	1.536	0.34 %	1.027	0.89 %
KDE, Silverman	4.123	0.00 %	2.028	0.00 %
KDE, adaptive kNN	5.304	0.00 %	3.999	0.00 %
memoriser, jitter 0 (<i>control</i>)	0.000	100 %	0.000	100 %
memoriser, jitter 0.02	0.399	9.0 %	0.207	99.9 %

MAGI does not memorise. Both ratios exceed 1, and both near-duplicate fractions are *below* real data’s own — MAGI reproduces training points less often than real crossings coincide with each other. Recall drives the ratio down, and no point on the memoriser curve approaches these values.

The metric is calibrated in both directions before being trusted: held-out real data reads as real (0.999/1.001, KS $p = 0.81/0.96$) and literal copies read as pure recall (0.000, 100 %). That two-sided validation is what makes the MAGI numbers interpretable.

CR at 1.027 is materially better than DM1.2 at 1.536: the CR model sits essentially where real data sits, while DM1.2 is over-dispersed by $\sim 54\%$ in nearest-neighbour distance — the opposite failure from memorisation, and consistent with its 5–11 % marginal defects. KDSOURCE is $2\text{--}5\times$ further off the manifold in both bandwidth modes, which is the trade behind its correlation win and its 15.45 % physically-impossible samples.

Controls that failed first, and were fixed

The KDE arm was originally introduced as “a memoriser by construction”. It is not: at Silverman bandwidth in 7-D the kernel is $\sim 4\times$ the nearest-neighbour spacing, so it over-smooths. A second control at jitter 0.02 was then judged against a hand-picked 50 % near-duplicate threshold, which its own displacement (0.053) could not reach past $\varepsilon = 0.033$. Both were replaced by a calibration curve. Neither error survived into a result, because the script refused to conclude while its negative control was unvalidated.

8 Cost — measured on DM1.2 and on SRON

The cost claim must not be an absolute wall-clock number, which depends on the machine. It must be a *ratio* with both sides measured on the same hardware, and it must be decomposed — because the naive figure is wrong by two orders of magnitude. DM1.2 was long the only case where both sides ran here, since the SRON reference ran on a cluster [E15]. As of 15 August 2026 SRON has been measured directly, by re-running *both* arms locally back to back (§8.1) [E18].

Step	wall	jobs	core-h	output
full simulation, 2×10^8 muons	12.71 h	8	101.67	999,358 crossings, 1,733 events
MAGI generation, 1.2×10^6	31 s	6	0.05	—
MAGI training, 3 seeds	486/620/743 s	1	0.51	—
MAGI injection, 1.2×10^6	9.74 h	6	58.46	2,019 events

The efficiency factor is not the speed-up. This is the trap, and it belongs in the paper explicitly because a referee will look for it.

Detector events per thrown muon	8.665×10^{-6}
Detector events per injected crossing	1.683×10^{-3}
Statistical amplification	194.2 × (ideal $1/\chi = 200.13$)
core-h per thrown muon	5.084×10^{-7}
core-h per injected crossing	4.872×10^{-5}
Transport cost ratio	95.8 ×
Net throughput gain	194.2/95.8 = 2.03 ×

Cross-checked by equivalent statistics: 1.2×10^6 crossings carry 2.402×10^8 equivalent muons, so MAGI delivers 4.11×10^6 equiv-muons/core-h against the full simulation's 1.97×10^6 — $2.09 \times$, consistent. A crossing costs $\sim 96 \times$ more to transport than a thrown muon *because it starts on the sphere, inside the assembly, in material and adjacent to the detector*, whereas most of the 2×10^8 thrown muons miss the apparatus and are killed almost immediately. MAGI removes the cheapest transport in the problem.

General form, and why DM1.2 is the pessimistic case.

$$\text{speed-up} = \frac{1}{\chi} \times \frac{c_{\text{primary}}}{c_{\text{crossing}}} \quad (3)$$

c_{primary} **here is the cost per primary of the *whole* run, outer and inner together** — $2 \times 10^8 \times 5.084 \times 10^{-7} = 101.7$ core-h, the full simulation. With that reading $\chi^{-1} c_{\text{primary}} / c_{\text{crossing}}$ is $C_{\text{total}} / C_{\text{inner}}$, and the equation returns the throughput gain directly. Substituting an *outer-only* cost per primary instead returns $C_{\text{outer}} / C_{\text{inner}}$, which is smaller by exactly one; see §8.1, where that is the difference between $2.33 \times$ and $3.33 \times$.

One full run decomposes into outer transport (primary $\rightarrow$ sphere, **eliminated** by MAGI) 53.0 core-h, 52 %; and inner transport (sphere $\rightarrow$ detector, **still paid**) 48.7 core-h, 48 %. The gain is set by how much material sits *outside* the CryoSphere relative to inside. DM1.2 is close to the worst case: a small laboratory cryostat with an $R = 95$ mm sphere enclosing most of the assembly. For a spacecraft primaries traverse metres of structure while the sphere-to-detector distance stays at centimetres, so the gain should be far larger — now measured on SRON, where the outer share rises from 52 % to 70 % (§8.1).

Methodological cross-check. The inner cost per crossing is measured twice, independently: 4.8731×10^{-5} core-h from the full-simulation arm (48.7 core-h / 999,358 crossings) and 4.8717×10^{-5} from the MAGI injection arm ($58.46 / 1.2 \times 10^6$). They agree to 0.03 %. The two-arm decomposition is therefore not an assumption; it is verified.

Reuse across design variants — where the speed-up lives. The decisive property is that the CryoSphere crossing population is determined *entirely by the geometry outside the sphere*. Design iteration changes the inside — detector layout, filters, readout electronics, shielding around the absorber — and leaves the outer structure untouched, so the training set stays valid across the

whole campaign and the full simulation is paid **once**, not once per variant. For N inner-design variants at reference-equivalent statistics: full simulation $N \times 101.7$ core-h; MAGI $102.2 + N \times 48.7$ core-h.

N variants	full sim	MAGI	speed-up
1	102	151	$0.67\times$
2	203	200	$1.02\times$
5	508	346	$1.47\times$
10	1,017	589	$1.73\times$
20	2,033	1,076	$1.89\times$
∞			$2.09\times$

Break-even at $N \approx 1.93$ — the second design variant. This is the same arithmetic as “ $1.93\times$ more statistics of the same geometry”, but it means something entirely different: nobody wants to re-run an identical geometry for twice the statistics, whereas testing a second inner configuration is what a design campaign does on day one. For a flight model, where the outer share exceeds DM1.2’s 52 %, the break-even approaches $N = 1$ — MAGI pays for itself on the *first* redesign.

8.1 SRON — the flight-like model, measured

Both arms re-run locally, back to back, on an idle machine at identical parallelism (6 jobs), in the geometry the arm-A reference used [E18]. Measuring them together is the point: the previously stored c_{crossing} came from a run whose machine load cannot be reconstructed, and a ratio of two numbers taken under different conditions is not a measurement.

Arm	wall $\times$ 6 jobs	core-h	output
1 — outer, 21,600,000 primaries	32.8 min	3.277	125,352 crossings
2 — inner, 639,996 crossings	71.9 min	7.192	—

$c_{\text{primary}} = 1.5170 \times 10^{-7}$ core-h (**outer only**), $c_{\text{crossing}} = 1.1237 \times 10^{-5}$ core-h. Locally measured $\chi = 5.8033 \times 10^{-3}$, i.e. $1/\chi = 172.3$, against the stored 6.084×10^{-3} ($1/\chi = 164.4$): the stored value is confirmed to 4.6 %, which is a genuine run-to-run difference, not Poisson (± 0.3 % on 125,352 crossings).

Correction to the run’s own output

`run_speedup.sh` prints “SPEED-UP = 2.33x”. That is $C_{\text{outer}}/C_{\text{inner}}$, not the speed-up: ARM 1 measures the *outer* cost per primary, whereas the DM1.2 formula takes the *full-run* cost per primary. The two differ by exactly one.

$$\text{speed-up} = 1 + \frac{1}{\chi} \frac{c_{\text{primary, outer}}}{c_{\text{crossing}}} = 1 + 2.326 = \mathbf{3.33\times}$$

The script’s own printed split confirms it independently: outer 70 %, inner 30 %, and $1/0.30 = 3.33$. With the stored χ , $3.22\times$. **Quote $3.33\times$ (or $3.22\times$), never $2.33\times$.**

	DM1.2 (lab)	SRON (flight-like)
outer share of one full run	52 %	70 %
speed-up, asymptotic	$2.09\times$	$3.33\times$
break-even, N variants	1.93	1.43

This is the prediction of §8 above, and it is now measured rather than asserted: more material outside the sphere, larger gain. SRON break-even ignores the training cost, which was not timed in this run; DM1.2’s 1.93 includes it (0.51 core-h, 0.5 % of one full run), so the SRON figure will move by a comparable and small amount.

Both numbers are lower bounds. The Building → Dome transport upstream is also eliminated by MAGI and is not counted in either arm. It cancels in the spectral comparison — reference and training set descend from the same chain — but it does *not* cancel here.

The claim that does not depend on compute at all. MAGI converts a **statistically exhausted** sample into an unlimited one. The ^{40}K training set holds 1,873,609 real crossings and more cannot be had without re-running an enormous campaign; MAGI generated 538,205,667 from it, a **287× amplification**, and the generated population closes at the detector at 1.118 ± 0.373 . No amount of compute buys that from the original sample.

The arithmetic is independent of *which* population was amplified, so the 287× stands as stated. But that population is the albedo set of §5, so the sentence to use is “MAGI amplifies an exhausted sample and the amplified population closes at detector level” — a statement about the method — and *not* “MAGI reproduces the ^{40}K background”. Until the September regeneration, DM1.2 and CR carry every physics claim.

9 Correction: the “seed systematic” is withdrawn

Withdrawn claim

Earlier versions of the internal documents reported that MAGI’s detector-relevant energy ratio spans 0.455–1.222 across training seeds, and recommended quoting a $\pm 40\%$ single-seed / $\pm 15\%$ ensembled systematic. **That spread is an artefact of the metric and is withdrawn.** It was never measured on a physical quantity.

The metric `energy_vs_impact_parameter` compares median generated to median real energy among detector-aimed crossings. On the CR source that statistic estimates the *species mix*, not the energy conditional.

At $b < 10$ mm the real population is [E12]: μ^- 48.008 % at median 828.373 MeV; γ 49.789 % at median 0.511 MeV; e^- 2.092 % at median 3.170 MeV; aggregate median 13.288 MeV. The two dominant populations are three orders of magnitude apart, split almost exactly 50/50, and *disjoint* (γ 99th percentile 30.488 MeV, μ^- 1st percentile 24.072 MeV). The 50th percentile of the combined sample therefore falls in the empty gap between them, and the aggregate median is close to a step function of the muon fraction.

Demonstration using the real data only. Reweighting the species mix of the real $b < 10$ mm crossings, with every energy value left untouched, so that the energy error is zero by construction [E12]:

μ^- reweight	μ^- fraction	aggregate median	ratio to truth
0.955	46.834 %	9.287 MeV	0.699
0.980	47.686 %	11.754 MeV	0.885
1.000	47.953 %	13.172 MeV	0.991
1.012	48.325 %	14.982 MeV	1.128
1.050	49.268 %	21.316 MeV	1.604

A $\pm 5\%$ mix shift swings the metric over 0.70–1.60. The measured per-seed muon reweights at $b < 10$ mm are 1.012/1.012/0.955 — precisely this range, and precisely the seeds that produced the reported spread.

The energy conditional, resolved per species [E12]:

Species	seed 42	seed 7	seed 13
γ	1.000–1.001	1.000–1.001	1.000–1.039
μ^-	0.956–1.003	1.043–1.064	0.947–0.976
e^-	0.680–1.303	0.805–1.005	0.555–0.615

(ranges over the cuts all-ingoing, $b < 50$, $b < 20$, $b < 10$ mm)

Gammas are reproduced *exactly* at every cut and every seed; muons within $\pm 6\%$. e^- scatters but is 1.9–2.1 % of the population with 639 real events at $b < 10$ mm — statistics-limited, not a demonstrated defect.

What is real. A $\pm 5\%$ seed variation in the species mix *conditional on direction* (μ^- reweight 1.012/1.012/0.955 at $b < 10$ mm, against global fractions correct to 0.3 %) [E12]. This is genuinely a joint-distribution failure that marginal validation cannot see — an order of magnitude smaller than claimed, and in a different channel. The same signature appears independently on DM1.2 (§3).

Consequence for §4. The §4b diagnostic in the validation note — “the model has invented an anti-correlation between energy and aiming” — rests on the same aggregate statistic and is **not currently supported**. Its marginals table, aimed-fraction column and counts are unaffected. The detector-level $R = 0.848 \pm 0.125$ never used the metric and stands.

10 v0.8.3: implemented, tested, not adopted

`energy_condition_geometry=True` plus `latent_dim 8 → 12`, trained at three seeds [E13]. It did not improve the target metric and **regressed the two lowest energy decades consistently across all three seeds**: 0.001–0.01 MeV from 0.880/0.921/0.995 to 0.677/0.715/0.738; 0.01–0.1 MeV from 1.686/1.296/1.767 to 2.216/1.714/1.979. That regression is measured on *marginals* and therefore survives the correction in §9; it remains a sufficient reason not to adopt v0.8.3.

Since §9 shows the energy conditional was not the broken term, `energy_condition_geometry` was aimed at a quantity that is not at fault, which explains why it did not help. The flag remains in `core/model.py` defaulting to `False`, so v0.8.2 is bit-identical and the experiment is resumable.

Superseded in part, 19/08. That reading stands on the marginals, but the runs behind it changed `latent_dim` as well as the flag and used the contaminated training set, so they never isolated the coupling. §7.3 then found a joint mismatch the marginals cannot see. Both facts point the same way: the flag deserves one controlled test against a metric sensitive to the joint, which is §11.

Revised priority for the next version.

1. Make `energy_vs_impact_parameter` species-resolved by default and refuse an aggregate on multi-species input.
2. Target the aimed species mix: condition the *type* draw on sampled geometry, rather than sampling from a global `type_probs`.
3. Measure the detector-level seed spread directly (in progress, §12).
4. More generated events at small b to settle whether the e^- scatter is real.

11 Head coupling, re-tested against the joint metric: closed, null

Launched 19 August 2026 09:37 CEST [E19]. §7.3 left the conversion deficit characterised but not mechanistically explained, and named `energy_condition_geometry` as the mechanism it calls for. This section records the test.

Result: null. Coupling does not close the gap

Scored 19 August 2026 19:56 CEST. Classifier AUC against 400,000 real crossings:

Arm	AUC	$c>0.8$ density ratio	Hole composition
Reference (Colab)	0.6604	28.7×	93 % gamma at 511 keV
Baseline (local)	0.6630	27.4×	95 % gamma at 511 keV
Coupled (local)	0.6568	25.0×	94 % gamma at 511 keV

$\Delta\text{AUC} = -0.0062$ ($0.6630 \rightarrow 0.6568$), against the $|d| \geq 0.03$ threshold set before scoring. **No meaningful change.** The conditional-independence edge between energy and geometry is not what the classifier is reading; the next candidates are the geometry heads themselves (Gaussian/unit-circle, no flow, unlike the energy head’s RQS flow) or the latent.

The reference-vs-baseline gap (0.6604 vs 0.6630 , $\Delta = 0.0026$) is the device/seed control this design bought: it bounds Colab-GPU/M1-CPU and seed-to-seed AUC variation well below the coupling effect, so the null above is not noise. The $c>0.8$ coverage hole persists at closely the same size and composition in all three arms — 25–29%, 93–95 % gamma at 511 keV — confirming it is a property of the coverage mechanism this architecture family shares, not of any one training run.

Why the v0.8.3 checkpoints could not answer this. The three v0.8.3 runs [E13] moved `latent_dim` $8 \rightarrow 12$ alongside the flag, and were trained on the *pre-ingoing-fix* CR set — their configs record `n_types` = 6 against the corrected set’s 4. They are therefore neither a controlled test of the coupling nor trained on the data the current result rests on, whichever metric they are scored by. The A/B below is the first controlled test the flag has had.

Design. `run_coupling_ab.py` builds the dataset *once* — features, gate targets, quantile transformers, split, conditioning, tf.data pipelines — and trains two models on that identical object: 40 epochs, seed 42, M1 CPU, 2,616,145 training rows. `energy_condition_geometry` is the only line that differs. The Colab reference checkpoint is re-scored alongside both, so device and seed variation between it and the two local models is measured rather than assumed. Scoring is classifier AUC against 400,000 real crossings: sensitive to the joint, and with negligible statistical error next to the ± 0.09 that R carries.

The implementation is sound. `core/model.py` factorises $p(\text{geom} \mid z)p(E \mid \text{geom}, z)$ with teacher forcing on true targets during training and re-decoding from *sampled* geometry at generation, so training and generation apply the same conditional. The geometry heads never see `geom_obs`, so nothing leaks into their own reconstruction term.

Third CR arm. [E20] The coupled checkpoint goes through the same detector chain as the baseline, in `cr_arm_coupled/` — a separate directory, so the reported campaign of §7.2 is untouched and stays reproducible from its own. The real arm is **not** re-run: `job_real_*` is reused verbatim, so all three arms share one reference exactly and the Geant4 cost is halved to three jobs. Generation seeds ($9200 + 53j$) and Geant4 seeds ($k = 3, 4, 5$) are the baseline MAGI arm’s, so the coupled arm differs from `job_magi_*` in exactly one thing: the checkpoint.

Because both MAGI arms share the real arm their R values are correlated, and the gap between their error bars is not the error on their difference. The reported statistic is therefore

$$Q = \frac{(\text{events/crossing})_{\text{coupled}}}{(\text{events/crossing})_{\text{baseline}}},$$

in which the real arm cancels entirely, and which is paired on the Geant4 side as well. $Q = 1$ means coupling changed nothing at the detector.

Result, 20/08/2026 02:27 CEST: Q confirms the AUC null

Band	R_{baseline}	R_{coupled}	$Q = R_{\text{coupled}}/R_{\text{baseline}}$
All deposits	0.920 ± 0.034	0.839 ± 0.031	0.911 ± 0.035
0.1–30 keV (X-IFU)	0.846 ± 0.088	0.906 ± 0.092	1.072 ± 0.114
1–20 keV	0.767 ± 0.087	0.847 ± 0.094	1.105 ± 0.131
1–7 keV (MIP)	0.807 ± 0.124	0.883 ± 0.133	1.094 ± 0.174

Headline band: $Q = 1.072 \pm 0.114$, $+0.6\sigma$ from unity. **Coupling did not move the detector rate.** The AUC $\rightarrow R$ link was the open question this arm existed to test; it holds in the direction that matters here, since a null on one metric and a moved value on the other would have been the informative outcome, and neither happened. Both the classifier and the detector agree the conditional-independence edge between energy and geometry does not drive the conversion deficit.

The production model stays `v0_8_2_CR_ingoinfix`. The memorisation test (§7.4), the classifier test (§7.3), the reweighting and arm A (§7.2) all remain keyed to it, unaffected by an arm that was always additive, never a proposed swap.

An AUC improvement is **not** licence to quote an improved R . AUC is a distributional distance on the crossing surface; R is an integrated detector rate after inner transport. That the first implies the second is the hypothesis under test, not a premise — which is the whole reason the third arm exists.

One limit to read the result against. The CR gate carries `zones = 2`: continuum plus a single pinned line at 511 keV. There is less for the coupling to express here than on a line-rich source, so a null on CR bounds the mechanism on *this* source rather than in general.

Status as this section stands. Closed. AUC: done, null (19/08 19:56). Third arm (Q): done, null (20/08 02:27 [E20]), landing inside the Thursday freeze with a day to spare. Both metrics agree: head coupling is not the mechanism behind the CR conversion deficit. The deficit itself remains open — see the working copy of this document on branch `v0.9.0-geometry-flow-heads` (not yet merged here) for the candidate this result redirected toward.

12 Result 4 — the detector-level seed test

Completed 14 August 2026, 22:31 CEST [E17]. Six jobs, two per checkpoint (seeds 42, 7, 13), 1,200,000 crossings each — verified exactly 1,200,000 records per file with no neutrinos — injected into `SRON_CCNwithXFDM_NoShield_FlowerCryoAC_fixed.gdml`, the same geometry as the arm-A reference [E14]. This is the measurement that had never been taken: every seed claim before it, withdrawn or not, was crossing-level.

12.1 Seed-to-seed dispersion

Counts per seed, and a χ^2 test of the hypothesis that the three differ only by Poisson fluctuation (2 d.o.f.; injected totals identical to one part in 2.4×10^6):

Channel	7	13	42	mean	χ^2	p
0.1–30 keV, Det 1	70	57	49	58.7	3.83	0.147
0.1–30 keV, all 6	129	103	90	107.3	7.35	0.025
1–20 keV, Det 1	66	55	44	55.0	4.40	0.111
1–20 keV, all 6	107	83	71	87.0	7.72	0.021
all deposits, Det 1	78	59	52	63.0	5.75	0.057
all deposits, all 6	990	857	827	891.3	16.89	2.2×10^{-4}

This reverses “no seed systematic is established”

In the highest-statistics channel the seeds are **not** consistent with Poisson: $p = 2.2 \times 10^{-4}$, an excess dispersion of **9.1 %** over the Poisson floor of 3.4 %. A detector-level seed systematic **does** exist. It is of order 10 %, not the withdrawn ± 40 %.

The ordering $7 > 13 > 42$ is preserved in *all six* channels, and the significance tracks the available statistics exactly as it should if the effect is real and roughly multiplicative.

Four caveats, all of which weaken the number.

1. **The channel was chosen after seeing the data.** The pre-registered sensitivity (§9 revision, 08:00) named the X-IFU band — ~ 62 Det-1 counts, ~ 112 over six — and called the run *insufficient* for a ± 5 % effect. It was: in that band $p = 0.025$. The power came from the all-deposits channel, which was not pre-registered. Six channels examined; the nesting makes a Bonferroni correction conservative, but even $6p = 1.3 \times 10^{-3}$ survives.
2. **Three seeds.** The 9.1 % is a dispersion estimated from 2 d.o.f.; its own 68 % interval spans roughly 5–18 %. Quote it as “of order 10 %”, never as 9.1 %.
3. **Nested channels are not independent measurements.** The six rows are subsets of the same events; only the last row carries real power.
4. Per-seed jobs share a checkpoint but differ in Geant4 transport seed, which is inside the Poisson null, not additional to it.

This is most likely the aimed-mix defect, seen downstream. A 10 % detector-level spread is the right size for the ± 5 % direction-conditional species-mix error already documented at crossing level (§9) and for the coherent 4 % deposited-energy deficit on DM1.2 (§3). It is *not* evidence for the withdrawn median- E -vs- b claim, which remains wrong for the reason given in §9: that estimator measures the species mix on a bimodal source. The correct reading is that the defect was real, an order of magnitude smaller than first claimed, and in a different channel — and it propagates to the detector.

12.2 Against the arm-A reference

Seed	MAGI/real	counts
7	0.915 ± 0.168	70
13	0.745 ± 0.144	57
42	0.641 ± 0.128	49
Pooled	0.767 ± 0.122	176

Reference: 51 counts / 1,600,000 injected real crossings = 3.1875×10^{-5} (± 14.0 %). The CR deficit is confirmed at a third measurement, consistent with $R = 0.848 \pm 0.125$ (§4) — but these are **not independent**: both are capped by the same 51-count reference, whose ± 14 % dominates the pooled error.

What this makes actionable

The precision on the CR deficit is limited by the *reference*, not by MAGI. 176 MAGI counts against 51 reference counts: generating more MAGI events cannot improve it. Pinning the 15–24% deficit requires more *real-crossing* reference statistics.

13 What may and may not be quoted

Quotable

- CR, SRON, normalization-free: $R = 0.848 \pm 0.125$.
- CR, SRON, spectral shape: KS $p = 0.20$, indistinguishable from RUN CRYOAC.
- CR, SRON, efficiency: 124.1 ± 15.2 over 6 detectors ($R = 0.76$).
- DM1.2, 3-seed ensemble: $R = 0.970 \pm 0.032$ overall, 0.999 ± 0.037 in the MIP band, 100 % crossing coverage.
- DM1.2 efficiency: 199.9 ± 7.4 in the MIP band against ideal 200.13 (174.8 ± 38.2 over 0.1–30 keV, reference-limited).
- Correlation matrix, CR crossings: mean $|\Delta\rho| = 0.015$ over the 10 feature pairs, max 0.042.
- Geometry marginals reproduced to 5–11 % RMS per bin (u_r 5.5 %, u_v 6.9 %, ϕ_r 6.0 %, ϕ_v 11.1 %), against a 0.44 % Poisson floor — a real, characterized defect.
- ^{40}K , BuildingModel: $R = 1.118 \pm 0.373$, KS $p = 0.60$ — **only** as a closure test of MAGI against its own training population, which is albedo (§5). Never as a ^{40}K background result.
- ^{40}K cost structure, measured twice independently: the true crossing efficiency for the reduced slab, unshielded BuildingModel is $\chi \approx 1.49 \times 10^{-4}$ (A/B, 17/08) against measurement 2's 1.3992×10^{-4} — agreeing to 6 % — with a decay/crossing cost ratio of 13.39. That gives a $\sim 9 \times 10^4 \times$ achievable speed-up for a ^{40}K surrogate in this geometry. *Achievable, not realised*: the present checkpoint was trained on the albedo population, so realising it needs the September retrain.
- Species fractions reproduced to 0.3 %; impact-parameter distribution to within 2–9 %.
- Cost, DM1.2: **194** $\times$ statistical amplification per transported particle (ideal 200.13) and **2.03** $\times$ net throughput, both measured on the same machine; 52 %/48 % outer/inner cost split; break-even at the 2nd design variant.
- Cost, SRON: **3.33** $\times$ speed-up ($3.22 \times$ at the stored χ), 70 %/30 % outer/inner split, break-even at $N = 1.43$ variants; a lower bound, Building $\rightarrow$ Dome excluded (§8.1).
- ^{40}K : **287** $\times$ amplification of an exhausted sample. The arithmetic is independent of which population was amplified, so the factor stands; phrase it as amplification *of a sample*, not as reproduction of the ^{40}K background (§5).
- **A detector-level seed systematic of order 10 %**, measured on three seeds ($p = 2.2 \times 10^{-4}$, §12), with the post-hoc channel choice disclosed. Quote as “of order 10 %”.
- **CR ingoing-cut prediction, confirmed out of sample** (§7): predicted $\times 1.268$ per primary from the training sets before either model was injected, measured **1.248 ± 0.047** — 0.4σ . All four bands agree within 1σ .
- CR against the full simulation, Detector 1, 0–20 keV: 0.802 ± 0.078 on the censored training set, **0.941 ± 0.088** on the corrected one.
- **MAGI does not memorise** (§7.4): nearest-neighbour ratio 1.027 (CR) and 1.536 (DM1.2), both above unity, with near-duplicate fractions *below* real data's own. Metric validated in both directions — held-out real reads 0.999, literal copies read 0.000 at 100 % near-duplicates.
- KDSOURCE is 2–5 $\times$ further off the data manifold than MAGI on the same metric (Silverman 2.03/4.12, adaptive kNN 4.00/5.30).
- CR, SRON, three-seed pooled: $R = 0.767 \pm 0.122$ — not independent of the 0.848 ± 0.125 above (shared 51-count reference).

Not quotable

- **A $\pm 40\%$ seed systematic.** Still withdrawn. The measured detector-level effect is of order 10% (§12) and does not rehabilitate the original claim or its estimator.
- **9.1% as a precise figure** — 2 d.o.f., interval $\approx 5\text{--}18\%$.
- Any aggregate median-E-vs- b number (§9).
- The ^{40}K 44-days comparison (0.049) as a MAGI result. The *reason* has changed: not the $57\times$ source-volume story, which is superseded, but the albedo selection of §5.
- **Any ^{40}K , Ra-226 or Th-232 number as a statement about radioactive-background physics.** All three training sets are 100.00% ingoing w.r.t. the buggy sphere centre and are back-scatter populations (§5). They support method claims — fit quality, amplification, detector-level closure — and nothing about the physics of those isotopes.
- The stored ^{40}K χ values 1.756×10^{-5} (ingoing) and 2.270×10^{-5} (all) as crossing efficiencies. They are albedo-selected rates, low by $\sim 8\times$; the measured efficiency is $\sim 1.49\times 10^{-4}$.
- The $\sim 9\times 10^4\times$ ^{40}K speed-up as an achieved MAGI result — it is the measured cost structure of the geometry, and the current checkpoint cannot deliver it.
- The DM1.2 0.1–30 keV band as a *spectral* comparison, or as an “X-IFU band” — 41 reference events, and DM127 is an anti-coincidence detector, not a TES absorber.
- Sub-percent agreement on the geometry marginals — they are 5–11%.
- Impact-parameter bins below $b \approx 5\text{ mm}$ at present sample sizes.
- Absolute rates from any DM1.2 **focused-source** run.
- **A $\sim 200\times$ speed-up.** 200.13 is the statistical amplification, not the compute saving; the measured throughput gain is $2.03\times$ (§8).
- **“2.33 $\times$ ” for SRON** — the figure `run_speedup.sh` prints. It is $C_{\text{outer}}/C_{\text{inner}}$; the speed-up is $3.33\times$ (§8.1).
- Any numeric cost extrapolation to Athena/X-IFU itself. SRON XFDM is flight-*like* and now measured, but it is not the Athena mass model.
- Absolute wall-clock timings as a cross-machine claim.
- **“The CR discrepancy is resolved.”** The improvement from 0.802 to 0.941 is $+0.139 \pm 0.117$, only 1.2σ , on 107 and 114 Detector-1 events. What is established is that the prediction held, not that the deficit is gone.
- Per-injected-crossing ratios as a flux claim. That normalisation divides χ out, and χ is what the ingoing fix changed (§7). Reading the wrong column once turned a 0.4σ confirmation into an apparent 4.2σ refutation.
- Absolute rates from the `cr_retest` run as a flux result — injection used `Flower_fixed` while the corrected training set was generated in `noShields_withCryoAC`. The mismatch is common to both arms and cancels in the ratio only.

14 Open items

1. **The SRON conversion deficit — measured, bounded, mechanism unlocalised.** Arm A (§7.2) gives $R = 0.846 \pm 0.088$ in the X-IFU band on 3.6×10^6 crossings per arm, statistically identical to the old checkpoint’s 0.848 ± 0.125 . Normalisation, aiming and species mix are all excluded by direct measurement. What is left is a few-percent species-dependent distortion of the joint distribution (e^- median energy $1.21\times$ high among aimed crossings; the wide forward cone 0.934). **This is the one open defect that matters for the paper.** Note that the two architectural fixes already tried in v0.8.3 — conditioning the energy head on sampled geometry, and widening the latent — were implemented, tested and *not* adopted (§10), so there is no evidence yet that this is a cheap fix. The classifier test (§7.3) now gives a metric sharp enough to re-judge them.
2. **Resolved 21/08/2026.** Retraining on the corrected set moves Detector 1 from 0.802 ± 0.078 to **0.963 ± 0.043** (§7), and the per-primary prediction held at 0.4σ . The improvement is

- +0.161 ± 0.089, **1.8 σ** — `cr_retest_morestat` [E22] took Detector-1 statistics from 114 to 566 events, exactly the gap this item named. “The SRON deficit is substantially addressed” is no longer superseded phrasing; it is what the number now supports.
3. **Ingoing-only filter** on both SRON training sets, with χ recomputed, then retrain — makes SRON consistent with DM1.2.
 4. **Detector-level seed spread** on CR (§12).
 5. **e^- behaviour at small b** — needs generation, not seeds.
 6. **Outer/inner core-hour split on a flight mass model.** DM1.2 is measured (§8); the Athena/X-IFU case — where the argument is strongest — is not, and is the one number that would let the cost claim be stated for a real mission.
 7. `GeneratedParticlesRecycled` did not fire during $\sim 107\times$ recycling — the warning path in `PrimaryGeneratorAction.cc` is unreachable. Silent recycling is exactly what it exists to catch.
 8. **DM1.2 fluorescence lines** — the training set supports continuum and geometry but is $\sim 9\times$ short for lines.
 9. **Regenerate the ^{40}K training set — September.** The existing set, and Ra-226 and Th-232 with it, are back-scatter populations selected by the buggy sphere centre (§5). A correct campaign needs the fixed `EventAction` centre (-507.66), and would at once make the $\sim 9\times 10^4\times$ speed-up realisable, restore a radioactivity physics result, and give the low-statistics ladder a case with a 44-day reference to score against. Deferred deliberately: a new training set is expensive and DM1.2 plus CR carry the paper without it.

15 Summary — the basis for the paper

The claim

MAGI is a conditional VAE that learns the phase-space distribution of particles crossing a closed surface around a cryogenic detector, so that the expensive transport from the outer mass model down to that surface is paid once and the learned distribution is resampled thereafter. The paper’s contribution is not that the marginals are reproduced — that is table stakes for a generative model — but that the surrogate is validated **downstream**: its output is pushed through the real Geant4 mass model and the resulting *detector* spectrum is compared with the full simulation. That closure is rare in the ML literature and is the strongest thing here.

What was measured, and on what

Three independent sources, two mass models, one code path:

Setup	Reference	Result
DM1.2 lab cryostat, μ	2×10^8 muons (iso)	$R = 0.970 \pm 0.032$; MIP 0.999 ± 0.037
SRON XFDM, cosmic rays	real recorded crossings	$R = 0.848 \pm 0.125$
SRON XFDM, cosmic rays	RUN CRYOAC	mean flux ratio 0.79; KS $p = 0.20$
^{40}K , BuildingModel [†]	real recorded crossings	$R = 1.118 \pm 0.373$; KS $p = 0.60$

[†] Closure against its own training population, which is albedo (§5). Both arms draw on the same biased set, so this is a method result, not a ^{40}K physics result. Ra-226 and Th-232 share the defect. **DM1.2 and CR carry every physics claim until September.**

DM1.2 is the cleanest: a 3-seed ensemble, 100 % crossing coverage after the overlap fix, and an efficiency factor of 200 ± 7 in the MIP band against an ideal $1/\chi = 200$. ^{40}K passes at matched N , but against its own albedo training population (§5) — a method result, not a physics one. CR sits 15–24 % low and is the one open discrepancy.

Fidelity of the generated population

Correlation residual mean $|\Delta\rho| = 0.015$ (CR), 0.013 (^{40}K), 0.018 (DM1.2); species fractions to 0.3 %; energy marginals within a few percent per decade. **Geometry marginals are reproduced to 5–11 % RMS per bin**, against a 0.44 % Poisson floor — a real, characterized limitation, and the number to quote rather than any claim of sub-percent geometric fidelity.

Cost — state it carefully

The efficiency factor is *statistical amplification*, not compute saving. On DM1.2, measured with both arms on the same machine: amplification $194\times$ (ideal 200.13), transport cost ratio $95.8\times$, net throughput gain **$2.03\times$** . One full run splits 52 % outer / 48 % inner; MAGI eliminates the outer only, so speed-up = $\chi^{-1} c_{\text{primary}}/c_{\text{crossing}}$.

On SRON, the flight-like model, both arms were re-run locally and the gain is larger: outer share 70 %, **speed-up $3.33\times$** , break-even at $N = 1.43$ variants. Note that `run_speedup.sh` prints $2.33\times$ for this, which is $C_{\text{outer}}/C_{\text{inner}}$ and not the speed-up (§8.1); both SRON and DM1.2 figures are lower bounds, since Building $\rightarrow$ Dome is excluded.

The argument that carries the paper is **reuse**: the crossing population is fixed by the geometry *outside* the sphere, so a design campaign that iterates the inner assembly pays the full simulation once. Break-even is the **second design variant** on both geometries. The prediction that a flight-like model would do better than DM1.2 is no longer an assertion — it is measured, 52 % $\rightarrow$ 70 % outer share. Separately, MAGI amplified an exhausted ^{40}K sample $287\times$ — $1.87 \times 10^6 \rightarrow 5.38 \times 10^8$ crossings — which no amount of compute can buy from the original sample. State it as amplification *of a sample*; that sample is albedo (§5).

Limitations to state in the paper

1. **The CR 15–24 % deficit is not fully explained.** 3.6 % is attributed to an inconsistent ingoing-particle cut; the remainder is uncharacterized since the energy $\times$ geometry diagnosis was withdrawn (§9).
2. **A direction-conditional species-mix error of $\sim 5\%$.** Global species fractions are right to 0.3 %, but conditioned on pointing at the detector the mix is off by 5–9 %. On DM1.2 this produces a coherent 4 % deficit in deposited energy, traced to incidence angle (median $\sec\theta$ 0.955 of truth).
3. **A detector-level seed systematic of order 10 %**, now measured (§12): three seeds, all-deposits channel, excess dispersion 9.1 % over a 3.4 % Poisson floor, $p = 2.2 \times 10^{-4}$, with the channel chosen post hoc and only 2 d.o.f. behind the estimate. The previously reported $\pm 40\%$ remains withdrawn — it was an artefact of a metric that measures the species mix on a bimodal source, and this result is an order of magnitude smaller and in a different channel. It is the right size to be the already-documented $\pm 5\%$ aimed-mix error propagating downstream.
4. **SRON and DM1.2 are on different ingoing conventions** — the SRON training sets contain 10.4 % (^{40}K : 22.5 %) outgoing crossings, DM1.2 none.
5. **Cost is now measured on two geometries**, DM1.2 ($2.03\times$, the pessimistic case) and SRON ($3.33\times$). Both exclude the Building $\rightarrow$ Dome transport and are therefore lower bounds. Neither is the Athena flight mass model.

A methodological point worth making

`energy_vs_impact_parameter` passed its own self-verification cleanly (1.000 on identical inputs, 0.050 on an injected anti-correlation) while being the wrong estimator for the population it was applied to. Self-verification proves a metric computes what it computes; it cannot reveal that the statistic is inappropriate. On a bimodal multi-species source the aggregate median is a knife-edge function of the species mix — a $\pm 5\%$ mix shift swings it over 0.70–1.60 with *zero* energy error.

This cost a development cycle and is worth stating positively: it tells anyone reusing the method which check to run, and it is not visible in any diagnostic the field would normally apply.

Release readiness

Against the seven-layer picture: layers 1, 2 and 5 are strong (5 unusually so); layer 3 has one hole (no classifier two-sample test) that is now known to be load-bearing, since the defect found here is exactly what such a test detects; layer 4 has no memorization test while the paper quotes a $287\times$ amplification; layer 6 lost its strongest entry with the seed withdrawal; layer 7 is measured on DM1.2 only. **Layer 0 — LICENSE, tag, DOI — is not started and blocks release.**

16 Provenance

Table 4: Evidence tags. All paths are local to the analysis machine.

Tag	Artifact
E1	S1GDML_DM1.2/allOutputGDML_DM1_2_iso.dat, allDSCryoSphere_DM1_2_iso.dat — the 2×10^8 -muon iso reference run.
E2	S1GDML_DM1.2/Analysis/MAGI_DM1_2_iso_comparison.ipynb, executed 13/08/2026 22:57; jobs in S1GDML_DM1.2/magi_iso_run/job0..5. Figures: MAGI_DM1_2_iso_{spectrum,crossings,aiming}.png.
E3	magi_iso_run/gen_s0..5.log — the “CryoSphere for reconstruction” line emitted by MAGI/scripts/generate_geant_source.py.
E4	Incidence-angle and aimed-species analysis over /Volumes/X10Pro/tmp/magi_dm12iso_s0..5.gntbin against allDSCryoSphere_DM1_2_iso.dat, 13/08/2026.
E5	S1GDML_DM1.2/CRYOSPHERE_OVERLAP_FIX.md; src/GDMLDetectorConstruction.cc (R 105→95, overlap check on); gpsMu_iso.mac.
E6	S1GDMLSRON_CryoSphereEmission/ab_run/ arm A vs the v0.8.2 1.85×10^7 run.
E7	S1GDMLSRON_CryoSphereEmission/Analysis/DataAnalysis_pkg.ipynb, sections <i>RUN CRYOAC vs MAGI</i> and <i>Efficiency factor per detector</i> ; docs/MAGI_validation_note.md §2–§3.
E8	docs/MAGI_validation_note.md §5; src/EventAction.cc ingoing test applied to the training files.
E9	k40_run/run_now.sh, k40_run/run_real_bm.sh, k40_run/compare_real_vs_magi.txt.
E10	Old-Simulations/Sim/S1GDMLSRON/gpsPotassium40.mac vs S1GDMLSRONBuilding/gpsPotassium40.mac.
E11	athena-build/outputGDML_K40_v082.dat EventId analysis; DataAnalysis_pkg.ipynb cell 4; MAGI_validation_note.md §7.6.
E12	MAGI/tools/species_resolved_evsb.py, MAGI/logs/species_evsb.log, 13/08/2026; real reference MAGI/TrainingData/alloutputDSCryoSphereCR.dat (3,285,347 crossings, neutrino-free).
E13	MAGI/tools/train_v083_cr_seed.py, MAGI/tools/compare_v082_v083.py, MAGI/docs/v0.8.3_seed_variance.md.
E14	S1GDMLSRON_CryoSphereEmission/cr_seed_run/run_cr_seeds.sh, launched 14/08/2026 07:54, completed 22:31.

Tag	Artifact
E18	S1GDMLSRON_CryoSphereEmission/speedup_run/run_speedup.sh, 15/08/2026 00:21, speedup.log. Both arms, 6 jobs, idle machine, SRON_CCNwithXFDM_NoShield_FlowerCryoAC_fixed.gdml. ARM 1 wall 1966 s, ARM 2 wall 4315 s. The script’s printed “SPEED-UP” line is $C_{\text{outer}}/C_{\text{inner}}$; add one (§8.1).
E17	Same run, seed-dispersion analysis: counts per seed as tabulated in §12, χ^2 on 2 d.o.f. against a Poisson null with variance equal to the mean. Injected totals 2,400,000/2,400,000/2,399,999. Reference arm A: 51 counts / 1,600,000 injected real crossings.
E16	MAGI/paper_figures/make_fig1.py, make_fig2.py, table1_validation.tex.
E19	MAGI/coupling_test/run_coupling_ab.py, score_coupling_ab.py, coupling_ab.log, launched 19/08/2026 09:37. Checkpoints trained_models/coupling_{baseline,coupled}; results coupling_ab.json, coupling_auc.json.
E20	S1GDMLSRON_CryoSphereEmission/cr_arm_coupled/ (run_cr_arm_coupled.sh, report_cr_arm_coupled.py), armed 19/08/2026 10:35, chained behind E19. Reuses cr_armAB/job_real_[0-2] verbatim as the shared reference. The reader was dry-run against cr_armAB first and reproduced 3,583,201/1570/203 and 3,590,382/1448/172 exactly.
E22	S1GDMLSRON_CryoSphereEmission/cr_retest_morestat/ (run_cr_retest_morestat.sh, report_cr_retest_morestat.py), launched 20/08/2026 19:04, 12 jobs in 3 batches of 4, completed 21/08 20:29. Reuses cr_retest/job{0,1,2} (the original “new”-arm jobs) verbatim as part of the pool; cr_retest/job{3,4,5} (“old” arm) untouched. paper_figures/make_fig_cr_vs_full.py regenerated against the full 15-job pool.
E15	Timings from run-directory timestamps and log headers: S1GDML_DM1.2/iso_run/iso_run.log (12/08 12:36:49 → 13/08 01:19:21, 8 jobs); magi_iso_run/magi_iso.log (13/08 13:12:42 → 22:57:20, 6 jobs); MAGI/tools/dm12_iso_seed{42,7,13}.log (486/620/743 s). Both Geant4 runs saturated the same 8-core machine (8 vs 6 jobs), so the per-core normalization carries a $\pm 10\text{--}20\%$ systematic.

Companion documents.

- S1GDMLSRON_CryoSphereEmission/docs/MAGI_validation_note.md — the authoritative running record, including superseded sections.
- MAGI/docs/v0.8.3_seed_variance.md — the withdrawn seed systematic and the revised plan.
- S1GDML_DM1.2/CRYOSPHERE_OVERLAP_FIX.md — the overlap investigation.